Reproducible Answer Selection over Large Language Model Ensembles, with an Exact Kemeny Reference and a Ground-Truth Audit

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Abstract

Choosing among sampled language-model answers requires a selection rule whose reproducibility and accuracy can be assessed separately. We formulate selection as an integer program combining pairwise judge scores, confidence, self-consistency and cost, and compute an exact Kemeny–Young reference. The quality-only ranking inherits a known 5-approximation guarantee for weighted ranking disagreement, not answer correctness. Under an independent binary-comparison model, we derive the exact accuracy gain over plurality voting for three candidates and enumerate break-even thresholds for larger finite pools. At 80% candidate accuracy, the three-candidate threshold is 80.35% mixed-pair judge accuracy. We audit one free-tier judge, scoring 60.5% on RewardBench, on six benchmarks comprising 532 distinct problems. Paired exact tests detect no homogeneous-pool advantage over self-consistency (10 versus 13 discordant problems of 532); the heterogeneous-pool selector performs worse (10 versus 32 of 402, $p < 0.001$). Offline ablations expose an objective-scale mismatch, while removing the cost term

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does not establish an advantage. These findings concern the tested judge and configuration, not stronger judges generally. The theoretical thresholds characterize binary Copeland selection under explicit assumptions; averaged soft judge scores do not directly estimate their accuracy parameter. The contribution combines a finite-pool benchmark for judge reliability with an auditable selection protocol, an exact ranking reference and stored outputs for offline comparisons.

Keywords: Artificial intelligence in OR, Large language models in OR, Integer programming, Group decisions and negotiations, Decision support systems

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1. Introduction

Sampling several answers can increase the probability that a language model produces a correct answer without ensuring that a selector finds it. Self-consistency (Wang et al., 2023) votes over extracted final answers; when all answers differ, its result depends on the tie rule. A language-model judge can distinguish such answers, but introduces another potentially unstable and inaccurate component (Atil et al., 2025; Landesberg, 2026). We study selection over fixed candidate pools, separating agreement with judge preferences from correctness against ground truth.

The operational decision arises after generation has already incurred its cost: a system must return one of the available responses. A correct answer may be present even when the most frequent answer is wrong, while a larger pool may add only repeated errors. Conversely, when several independent draws agree on the correct answer, replacing their vote with a noisy judge can discard useful evidence. The relevant comparison is therefore between

selection rules on the same candidates, rather than between separately generated responses. Checkable mathematics, science and knowledge questions provide ground truth for making that comparison.

Two distinct optimization questions follow. The first is which ordering best represents the judge's pairwise preferences. The second is which selected answer is correct. An exact ranking solution answers the first question but need not answer the second. OR-Select makes the selection objective explicit, while a Kemeny–Young solution supplies a ranking reference. The probabilistic analysis then asks a separate question: under a specified model of correct and incorrect answers, how accurate must binary comparisons be for their Copeland winner to outperform voting?

The paper makes three contributions. First, it derives an exact finite-pool condition under which binary pairwise judging with Copeland aggregation outperforms plurality voting: a closed form for $N = 3$, enumeration for $N \leq 7$, and extensions to pair-dependent judges and colliding wrong answers (Section 4 and Online Supplement). Second, it combines a composite selector, OR-Select, with an exact Kemeny–Young reference and stored judge outputs for offline comparison (Section 3.2). Third, it audits one free-tier judge on six ground-truth benchmarks using paired tests, recovery relative to available headroom and weight ablations (Sections 5–6).

Outcome and use case. The tested selector has no detected advantage over voting with homogeneous generation and performs worse in the pooled heterogeneous condition. This does not establish that stronger judges cannot help. The threshold analysis quantifies the reliability needed by a specified *binary* selector under its assumptions; it is not a universal deployment test for soft-score or composite selectors. For applications with checkable an-

swers, the practical contribution is an audit: retain candidates and judge outputs, compare selectors on the same problems, and distinguish ranking fit from answer accuracy before replacing voting.

An application need not adopt the particular default weights to use this protocol. It can store one candidate pool per problem, freeze the judge outputs, and compare alternative aggregation or cost policies without confounding them with new generations. Applications with a reliable final answer extractor can include plurality directly; open-ended applications need a separately justified grouping or evaluation rule. The distinction between these settings matters: exact textual disagreement is not itself evidence that responses differ in meaning or quality.

2. Related Work

Answer selection and judges. Universal self-consistency (Chen et al., 2023), PAIRS (Liu et al., 2024) and LLM-Blender (Jiang et al., 2023) replace or supplement voting with model evaluations. Verifier-based selection uses trained reward models (Cobbe et al., 2021; Lightman et al., 2024); our selector is training-free. Preference-graph ensembling and denoising also have recovery guarantees and response-selection applications (Hu et al., 2024); graph aggregation itself is therefore established prior work. Positional bias affects judges (Zheng et al., 2023; Wang et al., 2024a), as does self-preference (Panickssery et al., 2024). We average input orders (Tang et al., 2024). Errors can be shared (Kim et al., 2025); benchmark judge scores need not predict within-prompt selection quality (Landesberg, 2026). These findings motivate paired correctness evaluation.

These approaches also make different uses of additional information. Uni-

versal self-consistency asks a model to identify agreement between free-form responses; pairwise ranking evaluates relations between responses; a trained verifier estimates correctness from task-specific supervision. They are not interchangeable baselines. Our experiments compare a training-free judge and stored-output selectors, and do not estimate the performance of a separately trained verifier. The coverage-versus-selection distinction in inference-time scaling (Snell et al., 2025) motivates reporting both the availability of a correct answer and the fraction of that opportunity actually recovered.

Ranking and voting. The Kemeny–Young rule (Kemeny, 1959; Young & Levenglick, 1978) is NP-hard (Bartholdi et al., 1989); weighted-win ranking has a known 5-approximation for disagreement (Coppersmith et al., 2010). Its likelihood interpretation under a noisy ordering model (Young, 1988; Caragiannis et al., 2016) is related to, but different from, our two-class correctness model with fair same-class comparisons. Recent analyses study reward-based selection against voting (Di et al., 2026; Komiyama et al., 2026) and heterogeneous, correlated ensembles (Ai et al., 2026). In particular, Komiyama et al. (2026) report voting outperforming judges. Our analysis instead fixes a binary pairwise rule and derives its exact finite-pool accuracy difference from plurality. This identifies the comparison reliability needed at a given candidate accuracy, rather than inferring selection quality from a ranking objective alone.

The distinction from the closest inference-scaling analyses is specific. Di et al. (2026) combine answer frequencies and reward information in a minimax analysis of pass-at-budget selection. Komiyama et al. (2026) study asymptotic ensemble performance and optimized model weights, and also compare practical selectors. Ai et al. (2026) use higher-order information

for aggregation of heterogeneous, correlated predictions. Here the object is the finite tournament induced by binary comparisons: the three-candidate calculation is explicit, larger small pools are enumerated, and the empirical audit keeps candidate generation fixed when comparing decision rules. He et al. (2026) instead learn when to overturn a majority in three-agent debates, and also report a judge losing to voting. Their selective-intervention criterion concerns the precision of majority overrides, rather than the mixed-pair accuracy needed by a complete Copeland tournament. Negative judge results alone are not the distinction.

Operational-research context. Exact Kemeny methods in this journal address median rankings and winner computation (Amodio et al., 2016; Rico et al., 2023); ensemble optimization selects constituent models (De Bock et al., 2020; Aktaş et al., 2024). Here the decision is which already-generated answer to return, with an exact ranking reference and an offline audit. The distinction is between optimizing ensemble membership before prediction and optimizing the returned answer after observing a candidate pool. Established ranking and optimization components provide the reference machinery; the finite-pool correctness comparison and its accompanying audit address the answer-level decision.

Judgment aggregation and selection among forecasts provide a related OR perspective (Brown & Reade, 2019; Jaspersen, 2022; De Baets & Harvey, 2020): the selection mechanism is itself a decision object, not simply a final implementation detail. This perspective motivates examining the scale of the objective terms and testing the selected answer against ground truth, alongside the quality of the fitted preference ranking.

3. Method

3.1. Pipeline

Figure 1 separates generation, judge calls and selection.

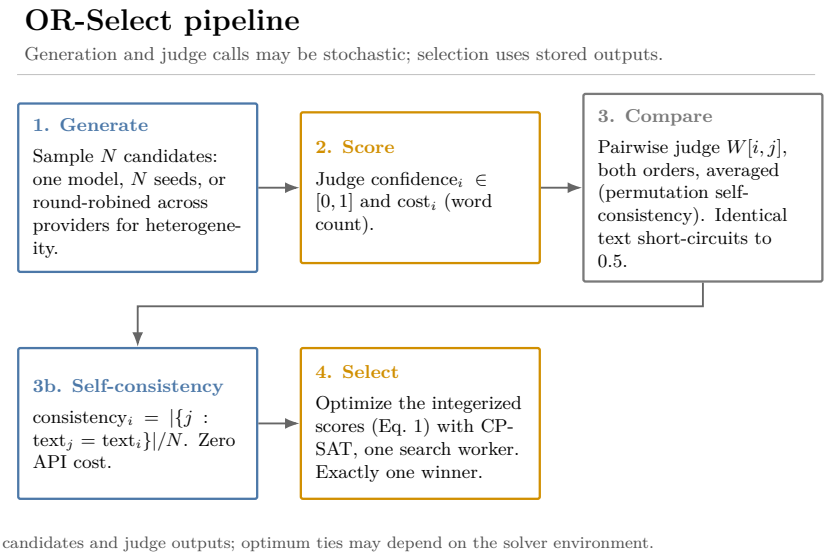


Figure 1: OR-Select. Generation and judge calls may be stochastic; aggregation and selection use their stored outputs. Self-consistency requires no API calls.

Sample N candidates from one model or a round-robin model pool. An invalid slot (for example, no extractable answer letter) is retried up to three times with varied seeds; persistent failures are reported as ungraded. Confidence $C_i \in [0, 1]$ comes from a judge call, cost c_i is word count, and consistency $K_i = |\{j : \text{text}_j = \text{text}_i\}|/N$ counts exact full-text matches, unlike final-answer voting.

For each unordered pair, average the judge's forward score and the complement of its reverse score to obtain $W[i, j] \in [0, 1]$; set $W[j, i] = 1 - W[i, j]$. This symmetrization mitigates order effects but does not guarantee unbi-

ased judging. Identical texts after trimming skip the judge and receive 0.5. At most $N(N-1)$ comparison calls are needed. The quality aggregate is $Q_i = (N-1)^{-1} \sum_{j \neq i} W[i,j]$.

The two agreement signals serve different purposes. Q_i measures how strongly the judge prefers candidate i to its competitors; K_i measures literal repetition of its complete response. Two derivations that end in the same number can therefore contribute two votes to the self-consistency baseline while each has $K_i = 1/N$ in the composite objective. Confidence is a separate pointwise judge assessment, not an observed correctness frequency. Keeping these quantities separate makes it possible to remove or reweight one without changing the saved comparisons.

3.2. Formal Model

Let x_i indicate selection and v_i denote validity. The selector solves

$$\max \sum_i x_i \left(w_1 Q_i + w_2 C_i + w_4 K_i - w_3 c_i \right) \tag{1}$$

$$\text{s.t.} \quad \sum_i x_i = 1, \qquad x_i \in \{0,1\}, \tag{2}$$

$$x_i \leq v_i \quad \text{if some candidate is valid,} \tag{3}$$

$$x_i = 0 \quad \text{if } Q_i < \text{min_quality} \quad \text{(optional).} \tag{4}$$

The default $(w_1, w_2, w_4, w_3) = (1, 0.2, 0.3, 0.01)$ was fixed before ground-truth runs, without tuning. A hundred-word difference therefore costs one full quality unit. Without additional constraints, this one-hot program is simply an argmax; integer programming provides a common implementation for the eligibility constraints. CP-SAT (Perron & Furnon, 2023) maximizes scores rounded after multiplication by 10^4, using one worker. Reproducibility

is conditional on the stored inputs and solver environment; this does not establish a canonical optimum across solver versions.

All terms are linear once the candidates have been scored. The validity constraint excludes invalid candidates if at least one valid candidate exists; an optional minimum-quality threshold excludes candidates below that threshold. Thus feasibility restrictions and preference weights have different roles. A low-cost response is not permitted to bypass an active validity restriction, while an eligible response can win through a combination of quality, confidence, consistency and cost. Reporting the component scales is consequently important even when the final optimization is computationally elementary.

Rows storing confidence and W permit offline weight changes. The default replay reproduces 729/729 extracted final answers; this comparison is at the final-answer level. A worked example is in the Online Supplement (Table S10).

3.3. Exact Kemeny–Young Baseline

For the ranking reference, let $y_{ij} = 1$ mean that i ranks above j. The Kemeny–Young program maximizes agreement with the same matrix:

$$\max \sum_{i \neq j} W[i,j]\, y_{ij} \tag{5}$$

$$\text{s.t.} \quad y_{ij} + y_{ji} = 1 \quad \forall\, i \neq j \tag{6}$$

$$y_{ij} + y_{jk} + y_{ki} \leq 2 \quad \forall \text{ distinct } i,j,k \tag{7}$$

$$y_{ij} \in \{0,1\} \tag{8}$$

Antisymmetry and the absence of 3-cycles enforce a linear order. The implementation uses $\binom{N}{2}$ binary variables, $O(N^3)$ constraints and coefficients

round$(10^4(W[i,j] - W[j,i]))$. A 200,000-conflict budget replaces a wall-clock limit; a non-optimality flag distinguishes feasible from proven-optimal solutions. All 854 stored reference solutions at $N \leq 6$ are flagged optimal for this integerized objective, not necessarily for unrounded weights.

The reference removes the composite selector's confidence, consistency and cost terms and uses all pairwise relations simultaneously. With cyclic preferences, its top-ranked candidate need not have the largest weighted win total. This permits an empirical distinction between losing information through aggregation and receiving unhelpful information from the judge. We compare returned answers, rather than treating a difference between rankings as an accuracy error. When two different candidates give the same extracted answer, they count as the same answer-level decision.

3.4. Approximation Guarantee

Proposition 3.1 (Weighted-win ranking)**.** *Assume nonnegative pair weights with* $W[i,j] + W[j,i] = 1$*. Ranking by decreasing* Q_i *approximates minimum weighted disagreement within a factor of 5. This factor is tight.*

The constructed matrix satisfies these conditions. This guarantee, from Coppersmith et al. (2010), covers the *full quality-only ranking*, not top-1 correctness, agreement with the Kemeny winner, or the composite objective (1).

4. When can a judge beat voting? Pairwise-accuracy thresholds

Model. Three candidates are independently correct with probability p, all correct candidates share the same final answer, and wrong answers are pairwise distinct. Plurality voting breaks answer ties uniformly. Conditional on the correctness labels, the judge's three *binary* comparisons are independent:

a mixed pair favors the correct candidate with probability q, and a same-class pair is a fair coin. Copeland selects the candidate with most wins, with an independent uniform tie-break. This quality-only model differs from the empirical selector's averaged soft scores and composite objective.

Theorem 4.1 (Pairwise-accuracy threshold)**.** *Under the model above, the accuracy of the Copeland selector minus the accuracy of plurality voting is*

$$\Delta(p,q) \;=\; p(1-p)\Big[(1-p)(2q-1)(q+1) \;-\; p\,(3-2q)(1-q)\Big].$$

Hence, for $p \in (0,1)$ and $q < 1$, the judge-based selector is strictly more accurate than voting if and only if $\frac{p}{1-p} < \frac{(2q-1)(q+1)}{(3-2q)(1-q)}$; at $q = 1$ it is always more accurate. In particular, (i) a judge at chance ($q = \frac{1}{2}$) selects with accuracy exactly p and loses to voting by $p^2(1-p)$; (ii) the unique break-even accuracy $q^*(p)$ increases strictly from $\frac{1}{2}$ (as $p \to 0$) to 1 (as $p \to 1$).

Proof. Conditional on exactly k correct candidates, write a_k for Copeland's correct-selection probability. Then $a_0 = 0$, $a_3 = 1$. With one correct candidate, winning both mixed comparisons succeeds; winning exactly one succeeds only through a three-way tie, giving $a_1 = q^2 + \frac{1}{3}q(1-q)$. With two correct candidates, winning both mixed comparisons succeeds; a split gives a correct selection with probability $\frac{1}{2} + \frac{1}{2} \cdot \frac{2}{3} = \frac{5}{6}$, hence $a_2 = q^2 + \frac{5}{3}q(1-q)$. Plurality's conditional probabilities are $(0, \frac{1}{3}, 1, 1)$. Weighting by $\mathrm{Bin}(3,p)$ gives

$$\Delta = 3p(1-p)^2(a_1 - \tfrac{1}{3}) + 3p^2(1-p)(a_2 - 1),$$

which simplifies to the stated expression. At $q = \frac{1}{2}$ it equals $-p^2(1-p)$. On $(\frac{1}{2}, 1)$ the ratio in the theorem increases continuously from zero to infinity: its numerator increases and its positive denominator decreases. Equating it

to $p/(1-p)$ establishes uniqueness, monotonicity and the endpoint limits of q^*. □

At $p = 0.8$, $q^* = 0.803515\ldots$ (Table 1). The formula gives both the sign and size of the gain under the model; `scripts/judge_threshold_theory.py` checks it by exact enumeration.

An explicit expression for the break-even point is

$$q^*(p) = \frac{2(1+2p)}{1+4p+\sqrt{9+8p-16p^2}}. \tag{9}$$

It follows by setting $\Delta = 0$ and rationalizing the relevant root of $2(1-2p)q^2 + (1+4p)q - (1+2p) = 0$; the expression also covers $p = 1/2$. For $p = 0.8$, plurality accuracy is 0.928. A judge with $q = 0.8$ yields 0.92672, just below voting, while $q = 0.9$ yields 0.96128, a gain of 3.328 percentage points. At $q = 1$, the selector finds a correct candidate whenever one exists, reaching 0.992. Thus the threshold is not a claim that judging cannot help: it identifies the reliability at which its benefit overtakes the information already supplied by answer repetition.

Table 1: Break-even binary judge accuracy q^* at which the Copeland selector ties plurality voting, as a function of the per-candidate accuracy p (Theorem 4.1, $N = 3$).

p	$q^*(p)$
0.2	0.565
0.4	0.631
0.5	0.667
0.6	0.706
0.7	0.750
0.8	0.804
0.9	0.875

Extensions and limits. The Online Supplement (Section S1) enumerates $N \leq 7$ and models wrong-answer collisions: each wrong candidate takes a shared attractor with probability ρ, otherwise a distinct answer. At $p = 0.8, \rho = 0$, the computed threshold rises from 0.804 for $N = 3$ to 0.945 for $N = 7$; at $\rho = 0.5$ it is approximately 0.82 for $N = 4, \ldots, 7$. A monotone-coupling result bounds a pair-dependent judge using its *minimum and maximum* mixed-pair accuracies, not their average (Corollary S1). The supplement also gives a large-N threshold lower bound and consistency regimes. Neither candidate independence nor uniform judge accuracy follows from a high aggregate benchmark score, and asymptotic voting consistency does not exclude finite-pool improvements by a stronger judge.

The finite-pool computation conditions on the number k of correct candidates. Let $a_{N,k}(q)$ denote the probability that a uniformly tie-broken Copeland winner is correct. Enumerating all binary tournament outcomes gives

$$A_J(N, p, q) = \sum_{k=0}^{N} \binom{N}{k} p^k (1-p)^{N-k} a_{N,k}(q). \tag{10}$$

With distinct wrong answers, plurality succeeds whenever $k \geq 2$ and with probability $1/N$ when $k = 1$, so

$$A_V(N, p) = 1 - (1-p)^N - (N-1)p(1-p)^{N-1}. \tag{11}$$

The computed break-even point equates these two accuracies. This is exhaustive enumeration of the finite comparison model, not an estimate from the benchmark observations; numerical root finding locates the crossing to the reported precision. The supplement compares it with an independent Monte Carlo calculation.

Table 2 illustrates the effect of wrong-answer collisions. Shared wrong answers strengthen the competing wrong vote, whereas with distinct wrong

answers two correct candidates suffice for plurality to succeed. The collision model changes the voting calculation; its binary judge still uses the stipulated mixed-pair accuracy. Its parameter ρ describes a common wrong answer, not arbitrary dependence between candidate correctness events.

Table 2: Finite-pool break-even binary accuracy $q^*(p, N, \rho)$, computed by exhaustive tournament enumeration. A wrong candidate takes a shared wrong answer with probability ρ, otherwise a distinct wrong answer.

	$\rho = 0$		$\rho = 0.5$	
N	$p = 0.5$	$p = 0.8$	$p = 0.5$	$p = 0.8$
3	0.667	0.804	0.625	0.782
4	0.759	0.891	0.675	0.814
5	0.800	0.919	0.682	0.824
6	0.825	0.935	0.679	0.819
7	0.841	0.945	0.673	0.821

The pair-dependent extension uses a coupling rather than substituting an average accuracy. Hold the correctness labels and comparison random numbers fixed, and increase one mixed-pair accuracy. The only possible change gives a correct candidate one additional win and a wrong candidate one fewer win; the correct fraction of the top-score set cannot decrease. Applying this change successively bounds selection accuracy between the uniform-$q_{\min}$ and uniform-$q_{\max}$ cases. This provides a conservative comparison when such bounds are available, without requiring every mixed pair to have exactly the same reliability.

5. Experimental Protocol

5.1. Setup

The fixed judge is NVIDIA Nemotron-3.5-Lightning on OpenRouter (`nvidia/nemotron-3.5-lightning:free`), with reasoning disabled and a 256-token cap; truncated judge replies raise errors. Homogeneous pools contain $N = 3$ Nemotron draws with distinct seeds. Heterogeneous pools contain $N = 6$ draws, initially two each from Nemotron, Poolside Laguna-S 2.1 and Cohere North-Mini-Code. After Laguna's withdrawal, Xing4.0-29B filled its slots (256 candidates); one early GPQA problem retains two LFM-2.5 candidates. This pool drift is reported, not treated as a fixed three-model experiment.

Generation uses a 32,768-token cap and varied-seed retries. A flagged second pass retries ungraded problems at 65,536 tokens, subject to model ceilings. Tables S8–S9 report coverage, pool composition and accuracy with and without these rescued problems.

The main references are self-consistency (plurality over extracted final answers, candidate-order ties), the *first sampled candidate* (not temperature-zero greedy decoding), and the Kemeny–Young top-ranked answer on the same matrix. Further offline selectors are listed in Section 6.4. Trained verifiers and universal self-consistency were not evaluated.

Judge calibration. RewardBench (Lambert et al., 2025) uses the same forward/backward judging path: its combined score must favor the human-preferred answer. The pre-registered sample takes the first seven rows of each of 23 subsets ($n = 161$), not a random sample. Overall accuracy averages category means; the supplement also reports a micro-average and its Wilson interval.

5.2. Evaluation tracks

Six homogeneous tracks comprise 532 distinct problems: B, 40 GSM8K test items (Cobbe et al., 2021); C, all 134 Level-5 items of MATH-500 (Hendrycks et al., 2021; Lightman et al., 2024); D, 30 AIME 2026 problems (Balunović et al., 2025); E, all 198 GPQA Diamond items (Rein et al., 2024); F, 30 AIME 2025 problems; and G, 100 MMLU-Pro test items (Wang et al., 2024b). B–E are repeated with heterogeneous pools on the same 402 problems, not additional independent benchmark items. MMLU-Pro samples five consecutive items at each of 20 evenly spaced offsets, covering 12 of 14 subjects. AIME 2025 may be contaminated because it predates the generator; it is a robustness check.

GSM8K uses the last number; MATH uses normalized exact match of the last `\boxed{}` answer; AIME uses the extracted integer; GPQA and MMLU-Pro use exact letters. String-based mathematics grading can reject equivalent expressions. The three-query open-ended pilot has no ground truth and is supplementary only (Section S6).

5.3. Metrics

We report graded/benchmark-total coverage and exact two-sided McNemar tests on paired discordant problems (Fagerland et al., 2013); pooled tests sum discordances within each generation condition. The test form was pre-registered, but the added tracks and six-track homogeneous pooling were not; pooled findings, per-track comparisons and ablations are exploratory, without multiplicity correction. A non-significant test is not equivalence.

The *floor* is mean candidate accuracy (uniform random selection), not a performance lower bound; the *oracle* is the proportion with at least one correct candidate. Recovery is $(\text{acc} - \text{floor})/(\text{oracle} - \text{floor})$ (cf. Landesberg,

2026), reported only when headroom is positive. Negative recovery means worse than a random pick.

6. Results

6.1. Judge validation: RewardBench

The fixed judge scores 60.5% overall, reaching 73.5% on reasoning pairs (36/49). Rotating judges yields 48.5%; this exploratory result informed retaining one fixed judge. Table 3 separates the categories. This calibration evaluates agreement with human-preferred responses; it is not the binary correct-versus-wrong parameter of the threshold model. Micro-averaging, length diagnostics and configuration history appear in Section S4.1 of the Online Supplement.

Table 3: Fixed-judge RewardBench accuracy by category ($n = 161$). The overall score is the mean of category accuracies.

Category	Rows	Accuracy
Reasoning (math-prm and six `hep-*` code subsets)	49	73.5%
Chat	35	68.6%
Chat Hard (adversarial LLMBar subsets)	42	42.9%
Safety	35	57.1%

6.2. Accuracy against the baselines

Tables 4–5 report accuracy and paired discordances. Across 532 homogeneous problems, OR-Select alone is correct on 10 and self-consistency alone on 13 ($p = 0.678$): a difference of -0.56 percentage points, without a detected advantage. Across 402 heterogeneous problems the counts are

10 versus 32 ($p < 0.001$), a -5.47-point difference. This loss comes from GSM8K, MATH and AIME 2026. Heterogeneous GPQA instead favors OR-Select numerically (10 versus 6 discordances, $p = 0.454$), so the pooled result is not uniform across tasks.

Table 4: Accuracy (%) on graded problems, by track and selector. '–': no Kemeny–Young comparator (GSM8K). Bold: best of the three practical selectors when they differ. Primed rows: heterogeneous generation.

Track	Graded/Total	First sample	Self-cons.	OR-Select	Kemeny–Young
B — GSM8K, homog.	40/40	100.0	100.0	100.0	–
B′ — GSM8K, heterog.	40/40	**100.0**	**100.0**	95.0	–
C — MATH L5, homog.	134/134	89.6	**91.0**	89.6	90.3
C′ — MATH L5, heterog.	134/134	88.1	**90.3**	79.1	79.9
D — AIME 2026, homog.	30/30	80.0	86.7	**90.0**	83.3
D′ — AIME 2026, heterog.	30/30	86.7	**93.3**	63.3	90.0
E — GPQA Diamond, homog.	198/198	75.3	**78.8**	**78.8**	78.3
E′ — GPQA Diamond, heterog.	198/198	74.7	77.3	**79.3**	72.7
F — AIME 2025, homog.	30/30	80.0	**83.3**	76.7	86.7
G — MMLU-Pro, homog.	100/100	85.0	**86.0**	**86.0**	83.0

6.3. Headroom and recovery

Table 6 distinguishes saturation from failed selection. Homogeneous GSM8K has no headroom; homogeneous GPQA has roughly ten percentage points, of which both selectors recover 16%. OR-Select falls below the random-pick floor on heterogeneous MATH and AIME 2026, whereas its numerical recovery advantage on heterogeneous GPQA is not statistically significant (Table 5).

6.4. Objective and selection diagnostics

On replayable problems, the within-pool cost-term spread has median 0.88, versus 0.20 for quality, and exceeds the quality spread in 653/729 cases.

Table 5: Paired comparison of OR-Select with each reference selector. Cells give (problems where OR-Select is right and the reference wrong)/(the reverse) and the exact two-sided McNemar p-value.

Track	vs. SC	p	vs. first sample	p	vs. Kemeny–Young	p
B — GSM8K, homog.	0/0	–	0/0	–	–	–
B′ — GSM8K, heterog.	0/2	0.500	0/2	0.500	–	–
C — MATH L5, homog.	0/2	0.500	1/1	1.000	1/2	1.000
C′ — MATH L5, heterog.	0/15	<0.001	4/16	0.012	7/8	1.000
D — AIME 2026, homog.	1/0	1.000	3/0	0.250	2/0	0.500
D′ — AIME 2026, heterog.	0/9	0.004	0/7	0.016	1/9	0.021
E — GPQA Diamond, homog.	6/6	1.000	13/6	0.167	9/8	1.000
E′ — GPQA Diamond, heterog.	10/6	0.454	20/11	0.150	24/11	0.041
F — AIME 2025, homog.	1/3	0.625	1/2	1.000	1/4	0.375
G — MMLU-Pro, homog.	2/2	1.000	3/2	1.000	5/2	0.453

Table 6: Headroom and recovery on graded problems. Floor: mean candidate accuracy. Oracle: some candidate correct. Recovery (%): (acc − floor)/(oracle − floor); negative means below a random pick.

Track	Graded	Floor	SC	OR	Oracle	Rec. SC	Rec. OR
B — GSM8K, homog.	40	99.2	100.0	100.0	100.0	100	100
B′ — GSM8K, heterog.	40	97.5	100.0	95.0	100.0	100	-100
C — MATH L5, homog.	134	89.8	91.0	89.6	91.8	62	-12
C′ — MATH L5, heterog.	134	81.8	90.3	79.1	91.8	85	-27
D — AIME 2026, homog.	30	85.6	86.7	90.0	90.0	25	100
D′ — AIME 2026, heterog.	30	76.7	93.3	63.3	96.7	83	-67
E — GPQA Diamond, homog.	198	77.1	78.8	78.8	87.4	16	16
E′ — GPQA Diamond, heterog.	198	71.8	77.3	79.3	92.9	26	35
F — AIME 2025, homog.	30	81.1	83.3	76.7	93.3	18	-36
G — MMLU-Pro, homog.	100	85.0	86.0	86.0	92.0	14	14

This exposes a scale mismatch, not a proof that cost determines every winner: confidence, consistency, eligibility and score gaps also matter.

Table 7 compares the default, no-cost, quality-only and plurality-plus-quality variants on identical replayable subsets. In heterogeneous pools, removing cost yields 78.6% accuracy and quality-only selection 78.1%, against voting's 84.9% ($n = 397$; exact p <0.001 and <0.001, respectively). Thus the cost term is not the only limitation. These post-hoc results do not prove that the judge contains no useful correctness information.

OR-Select picks Cohere candidates in 75.6% of heterogeneous problems, although their overall accuracy (83.9%) is similar to Nemotron's (82.5%). This is a property of the *composite selector*, not an isolated judge preference; length and other terms are confounded (Table S6).

Table 7: Offline replay of the selection step on stored judge outputs (accuracy %, problems whose judge outputs were stored). SC: stored self-consistency result on the same problems.

Track	Replayable	SC	Default	No cost	Quality only	SC+quality
B′ GSM8K, heterog.	38/40	100.0	94.7	100.0	100.0	100.0
C MATH L5, homog.	25/134	84.0	84.0	84.0	84.0	84.0
C′ MATH L5, heterog.	133/134	90.2	78.9	82.7	82.7	87.2
D AIME 2026, homog.	9/30	66.7	66.7	66.7	66.7	66.7
D′ AIME 2026, heterog.	29/30	93.1	62.1	75.9	75.9	96.6
E GPQA Diamond, homog.	168/198	77.4	78.0	78.0	78.0	76.2
E′ GPQA Diamond, heterog.	197/198	77.2	79.2	72.1	71.1	75.6
F AIME 2025, homog.	30/30	83.3	76.7	83.3	83.3	83.3
G MMLU-Pro, homog.	100/100	86.0	86.0	84.0	84.0	86.0

Additional selectors. Shortest/longest, token-Jaccard minimum-Bayes-risk, pointwise confidence, confidence-weighted voting, a knockout tournament and Bradley–Terry selection establish no significant improvement over plu-

rality in the exploratory comparisons (Table S5). Minimum-Bayes-risk is numerically ahead in some pooled comparisons, without a significant advantage.

Soft-score diagnostics. The mean stored weight assigned to the correct candidate on mixed pairs is 0.49–0.56 on tracks with at least 70 such pairs. These are *soft-score means*, not frequencies of correct binary decisions. They cannot be substituted for q in Theorem 4.1; neither can an average establish the pairwise bounds of Corollary S1. Table S7 retains these descriptive measurements and model sensitivity calculations, with their sampling limitations.

6.5. Ranking reference and coverage

OR-Select and the integerized Kemeny–Young reference return different final answers on 139/854 problems. OR-Select alone is correct on 50 and Kemeny–Young alone on 44 (Table 5); the exact ranking reference shows no detected accuracy advantage. Ranking optimality does not guarantee correctness.

Coverage. Every problem is ultimately graded (Table S8). Including the higher-cap rescued problems lowers selector accuracy by 0.3–15 points depending on track, without reversing the direction of the comparisons (Table S9). These results describe the retry-conditioned generation protocol, not unrestricted independent draws.

7. Discussion

Interpretation. The audit supports evaluating the entire decision rule on fixed candidate pools: a ranking objective can fit judge preferences without improving correctness, and an unnormalized cost term can overshadow

a bounded quality score. Stored outputs make these distinctions inspectable without new API calls. The theory separately quantifies the trade-off for independent binary comparisons. Connecting it to soft-score selection requires assumptions about score distributions and dependence, not merely a mean score.

Scope. Only one free-tier fixed judge is evaluated in the main audit. Stronger judges, trained verifiers and other selection architectures may perform differently. Small discordant counts limit precision; the replay subset is not randomly sampled, and mixed-pair measurements are sparse on several tracks. Candidate-order voting ties can favor particular generators in the round-robin pool. The reported results describe the generation, extraction and retry protocols of Section 5. The theory assumes independent candidate correctness and conditionally independent comparisons; the wrong-answer collision model does not model general correlated errors.

8. Conclusion

An exact finite-N calculation specifies when binary Copeland selection beats plurality under a stylized model. A separate six-benchmark audit finds no detected homogeneous-pool advantage for the tested free-tier judge and a pooled heterogeneous-pool loss. The exact Kemeny reference and offline ablations distinguish ranking fit, objective design and answer accuracy. Together, the finite-pool calculation and stored-output audit separate the reliability needed by a specified selection rule from the performance observed for a particular judge configuration.

Data and code availability

Code, tables, pre-registration documents and public run files are archived on Zenodo (Dressino, 2026). Development repository: `https://github.com/DressPD/or-select`. The public run archive excludes GPQA records containing password-gated question text; access to that benchmark follows its distribution conditions. The analysis scripts regenerate tables from stored runs and include the GPQA generation workflow.

CRediT authorship contribution statement

Aris Dressino: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation the author used Claude (Anthropic; Claude Code with Claude Sonnet 5.5) for analysis code, drafting, bibliographic checks and derivation and numerical verification of the theoretical results, and OpenCode (GPT-6.1 Sol) for critical review and editorial compression. The models served through OpenRouter are experimental subjects, not writing tools.

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